Page 2: The Human Eye – Structure and Function

The human eye is one of the most valuable and sensitive sense organs. It acts like a camera, enabling us to see the colourful world around us.

Key Parts of the Eye and Their Functions:

1. **Cornea:** The transparent spherical membrane covering the front of the eye. Light enters the eye through this thin, clear layer. It is the first and main refractive surface, providing most of the eye's focusing power.
2. **Iris:** The coloured part of the eye between the cornea and the lens. It is a muscular diaphragm that controls the size of the pupil.
3. **Pupil:** The dark, circular opening in the center of the iris. It appears black because light enters it and is absorbed. The pupil regulates the amount of light entering the eye, just like the aperture of a camera.
4. **Crystalline Lens:** A transparent, biconvex lens located behind the iris. It provides fine-tuning of focus to form a sharp image on the retina.
5. **Ciliary Muscles:** These muscles hold the lens and change its shape (make it thicker or thinner) to adjust the focal length. This process is called **accommodation**.
6. **Retina:** The light-sensitive screen at the back of the eye. It contains millions of light-sensitive cells called **rods** (sensitive to light intensity) and **cones** (sensitive to colour). The retina converts the optical image into electrical signals.
7. **Optic Nerve:** This nerve carries the electrical signals from the retina to the brain, where the image is interpreted.
8. **Vitreous Humour:** A clear, gel-like substance filling the space between the lens and the retina, maintaining the shape of the eye.

Page 3: Power of Accommodation and Defects of Vision

Power of Accommodation

The ability of the eye lens to adjust its focal length by changing its shape (via the ciliary muscles) to form clear images of objects at different distances on the retina is called the **power of accommodation**.

- **For distant objects (at infinity):** Ciliary muscles relax, the lens becomes thin (less curved), and its focal length increases.
- **For nearby objects:** Ciliary muscles contract, the lens becomes thick (more curved), and its focal length decreases.

Near Point and Far Point

- **Far Point:** The farthest point an eye can see clearly without strain. For a normal eye, it is **infinity**.
- **Near Point:** The closest point an eye can see clearly without strain. For a normal eye, it is **25 cm** (the least distance of distinct vision).

Defects of Vision and Their Correction:

Defect	Cause	Correction	
Myopia (Short-sightedness)	A person can see nearby objects clearly but distant objects appear blurry.	**Cause:** Either the eyeball is too long or the lens is too curved, causing the image to form **in front of** the retina.	Corrected by using a **concave lens** of appropriate power. This lens diverges the light rays slightly so they focus exactly on the retina.
Hypermetropia (Long-sightedness)	A person can see distant objects clearly but nearby objects appear blurry.	**Cause:** Either the eyeball is too short or the lens is too flat, causing the image to form **behind** the retina.	Corrected by using a **convex lens** of appropriate power. This lens converges the light rays more so they focus exactly on the retina.
Presbyopia	Gradual loss of the power of accommodation due to aging. The near point recedes.	**Cause:** Weakening of ciliary muscles and loss of flexibility of the lens.	Corrected by using **bifocal lenses**. The upper part has a concave lens (for distance) and the lower part has a convex lens (for reading).

Page 4: Refraction Through a Prism – Bending of Light

A prism is a transparent optical element with flat, polished surfaces that refract light. Typically, it is a triangular glass prism.

What happens when light passes through a prism?
Unlike a rectangular glass slab (which causes lateral displacement but the emergent ray is parallel to the incident ray), a prism bends light towards its base.

The Process:

1. **First Refraction (Air to Glass):** When a light ray enters the prism, it bends **towards the normal** because it is going from a rarer (air) to a denser (glass) medium.
2. **Second Refraction (Glass to Air):** When the ray exits the prism, it bends **away from the normal** because it is going from a denser to a rarer medium.

Because the two faces of the prism are inclined at an angle (the **angle of the prism**), the overall effect is that the emergent ray bends significantly away from the original path of the incident ray.

Important Terms:

- **Angle of Deviation (∠D):** The angle between the incident ray and the emergent ray. It is the total angle through which the prism bends the light ray.
- **Angle of Prism (∠A):** The angle between the two refracting faces of the prism.

Page 5: Dispersion of White Light by a Prism

One of the most beautiful experiments in physics was performed by **Sir Isaac Newton**. He showed that white light is not a single colour, but a mixture of seven colours.

What is Dispersion?
The splitting of white light into its constituent colours (the spectrum) when it passes through a prism is called **dispersion**.

Why does dispersion happen?
Dispersion occurs because different colours of light travel at different speeds in a glass prism. Since the refractive index of glass is slightly different for each colour, each colour bends by a different amount when passing through the prism.

- **Violet light** slows down the most, so it bends the **most** (has the highest refractive index for glass).
- **Red light** slows down the least, so it bends the **least** (has the lowest refractive index for glass).

The other colours (Indigo, Blue, Green, Yellow, Orange) bend by intermediate amounts between red and violet.

The VIBGYOR Spectrum:
The ordered sequence of colours we see is:
Violet, **I**ndigo, **B**lue, **G**reen, **Y**ellow, **O**range, **R**ed.
(Red deviates the least, Violet deviates the most).

Newton's Famous Experiment:
Newton used two prisms to prove that the prism does not "create" colours, but merely separates them.

- He used the first prism to disperse white light into a spectrum.
- He then placed a second prism upside down to recombine the spectrum. The second prism reconstituted the coloured rays back into white light. This proved that white light is composed of seven colours.

Page 6: The Rainbow – A Natural Spectrum

A rainbow is a natural spectrum of light that appears in the sky when the sun shines during or after rain. It is caused by the **dispersion**, **refraction**, and **total internal reflection** of sunlight by tiny, suspended raindrops.

The Process (Step-by-Step):

1. **Refraction (Entry):** When a ray of sunlight enters a spherical raindrop, it slows down and bends (refracts). As it enters, the white light is dispersed into its seven constituent colours.
2. **Total Internal Reflection (Inside):** The dispersed rays then strike the inner surface of the raindrop. If the angle of incidence is greater than the critical angle, the rays undergo **total internal reflection**. This is why the light is reflected back instead of escaping.
3. **Refraction (Exit):** The reflected rays exit the raindrop. As they go from water to air, they refract again, further increasing the separation of colours.

Why do we see an arc?
A rainbow is actually a full circle, but we usually see only an arc because the ground cuts off the bottom half. To see a rainbow, the sun must be behind you, and the rain in front of you. The centre of the rainbow arc is always opposite the sun.

Order of Colours:

- **Primary Rainbow:** Formed by one internal reflection. Red is on the **outer** edge, Violet on the **inner** edge.
- **Secondary Rainbow:** Sometimes a fainter, secondary rainbow is seen above the primary one. It is formed by two internal reflections. Its colour order is **reversed** (Red on the inside, Violet on the outside).

Page 7: Atmospheric Refraction

The Earth's atmosphere is made of layers of air with different densities and temperatures. This means the refractive index of air changes with altitude (it is slightly higher near the ground). As light passes through these layers, it bends gradually. This phenomenon is called **atmospheric refraction**.

Effects of Atmospheric Refraction:

1. **Twinkling of Stars:** This is not an intrinsic property of stars, but an effect of our atmosphere.
    - Starlight, entering the Earth's atmosphere, undergoes continuous refraction due to the changing refractive index of air layers.
    - The apparent position of the star keeps changing slightly.
    - The amount of light entering our eyes also fluctuates.
    - This makes the star appear to twinkle.
2. **Planets do not twinkle:** Planets are much closer to Earth and appear as extended sources (a collection of many point sources). The fluctuations in light from different points of the planet average out, so the overall intensity remains constant, and they do not appear to twinkle.
3. **Advance Sunrise and Delayed Sunset:** The sun is actually visible about **2 minutes** before it rises above the horizon in the morning and remains visible for about **2 minutes** after it sets below the horizon in the evening.
    - This happens because light from the sun, when it is just below the horizon, bends due to atmospheric refraction and reaches our eyes, making the sun appear higher in the sky than its actual position.

Page 8: Scattering of Light

When light passes through a medium containing particles (like dust, smoke, or gas molecules), a part of it is absorbed and then re-emitted in all directions. This phenomenon is called **scattering of light**.

Tyndall Effect:
The scattering of light by colloidal particles (particles suspended in a medium) is known as the Tyndall Effect.

- *Example:* The blue colour of the sky, the visibility of a light beam in a dark room (due to dust particles), or the headlight beams of a car in fog.

Why is the colour of scattered light different?
The amount of scattering depends on the size of the scattering particle and the wavelength of light.

Rayleigh's Law of Scattering:
According to Lord Rayleigh, the intensity of scattered light is **inversely proportional to the fourth power of the wavelength** of light.

Scattering $\propto 1 / \lambda^4$

This means:

- Shorter wavelengths (Violet, Blue) are scattered much more than longer wavelengths (Red, Orange).

Page 9: Why is the Sky Blue? Why are Sunsets Red?

1. Why is the Sky Blue?

- Sunlight, which is white light, has to travel through a long path in the Earth's atmosphere to reach our eyes.
- The atmosphere contains tiny gas molecules and particles. These particles are much smaller than the wavelength of visible light.
- According to Rayleigh's Law, **blue colour** (shorter wavelength) is scattered the **most** (about 10 times more than red) by these molecules.
- This scattered blue light enters our eyes from all directions, making the entire sky appear blue.

2. Why are Sunsets and Sunrises Red?

- During sunrise or sunset, the sun is near the horizon. The light from the sun has to travel a much **longer distance** through the atmosphere compared to midday.

- Most of the shorter wavelengths (blue, violet) are scattered away and lost from our line of sight by the time the light reaches us.
- The longer wavelengths (red, orange) are scattered the least. They travel almost straight through the atmosphere without much scattering.
- As a result, the light that reaches our eyes is predominantly red and orange, giving the beautiful reddish hue to the sun and the sky around it at sunrise and sunset.

Page 10: Summary and Revision Points

Phenomenon	Cause	Key Principle / Law
Power of Accommodation	Changing shape of eye lens via ciliary muscles.	Focal length adjustment for near/distant objects.
Myopia (Correction)	Image forms **in front** of retina.	Corrected by **Concave Lens**.
Hypermetropia (Correction)	Image forms **behind** retina.	Corrected by **Convex Lens**.
Refraction through Prism	Light bends twice; emergent ray bends towards base.	Angle of Deviation (D) depends on prism angle and refractive index.
Dispersion	Splitting of white light into 7 colours.	Different colours have different speeds/wavelengths in a medium. (VIBGYOR)
Rainbow Formation	Sunlight + Raindrops.	**Refraction, Dispersion, and Total Internal Reflection** inside raindrops.
Twinkling of Stars	Atmospheric Refraction.	Continuous bending of starlight due to changing air density.
Advanced Sunrise	Atmospheric Refraction.	Light bends downwards, making sun appear higher than it is.

Phenomenon	Cause	Key Principle / Law
Blue Colour of Sky	Scattering of Light.	**Shorter wavelengths (Blue)** are scattered more by atmospheric particles (Rayleigh's Law).
Red Colour of Sunset	Scattering of Light.	**Longer wavelengths (Red)** are scattered least; they travel through longer atmospheric paths.

Final Tip for Board Exams:
When explaining phenomena like "Why the sky is blue" or "Why a sunset is red," always mention **Rayleigh's Law of Scattering** and the concept of **wavelength**. For eye defects, clearly state the **cause** (where the image is formed) before stating the **correction** (which lens is used). Draw neat, labeled diagrams whenever possible.